Plan Presentation Speech Script (English)

Presenter: Qingpeng | Date: 7.1 | Duration: ~8 minutes

Slide 1: Title & Scope (~0:50)

Good morning everyone, and thank you for being here.

My thesis topic is the evaluation of RPCA-enhanced diffeomorphic registration for cardiac cine MRI motion tracking. Before introducing the details, I want to clarify the scope of this work. I am not proposing a brand-new registration algorithm. Instead, I focus on evaluation innovation. Specifically, I want to answer one core question in a rigorous way: after integrating RPCA into a diffeomorphic registration framework, do we obtain a truly meaningful and statistically significant improvement in motion tracking performance?

This means my contribution is not only to compare methods numerically, but to build a reliable evaluation protocol that can support stronger conclusions.

Slide 2: Background & Aim (~1:30)

Let me start with the background. Cardiac cine MRI is widely used to assess cardiac function, and motion tracking is an essential step for downstream analysis such as strain estimation. However, many existing studies rely on limited evaluation, often emphasizing only one metric, such as Dice.

The problem is that one metric cannot represent all aspects of registration quality. A method may show good overlap but still produce unrealistic deformation fields or unstable temporal behavior. Therefore, my plan has three aims.

Aim one is to build a multi-dimensional metric framework covering accuracy, topology, temporal consistency, low-rank behavior, and efficiency. Aim two is to apply paired statistical significance testing so that conclusions are statistically grounded rather than descriptive. Aim three is to provide a clear and balanced conclusion on whether DiffIR plus RPCA truly outperforms DiffIR under a controlled and reproducible setup.

Slide 3: Data & Experimental Setup (~1:40)

For data, I use the ACDC dataset as the primary benchmark. I choose it because it is public, widely used, and allows comparisons with prior studies.

I evaluate both pairwise and groupwise settings. Pairwise evaluation tests basic registration capability, while groupwise evaluation is closer to the true motion tracking scenario where temporal consistency matters.

The preprocessing pipeline includes resampling, intensity normalization, and region-of-interest cropping. To avoid data leakage, I use patient-level splitting, so images from the same patient do not appear in both training and testing subsets.

For method comparison, I keep two tracks under unified conditions: baseline DiffIR and proposed DiffIR plus RPCA. This controlled setup is important, because it ensures that any observed performance difference is more likely due to method design rather than inconsistent preprocessing or parameter settings.

Slide 4: Core Plan – Analysis (~2:40)

This slide is the core of my thesis plan.

First, the multi-dimensional metric framework. I evaluate accuracy using Dice, HD95, and ASSD. Dice captures overlap quality, while HD95 and ASSD better reflect boundary behavior.

Second, I evaluate deformation plausibility using the folding ratio, defined as the percentage of locations where the Jacobian determinant is less than or equal to zero. This indicates topology violations and is critical in medical registration.

Third, I evaluate temporal consistency by measuring frame-to-frame smoothness of the deformation fields. Since this is a motion tracking task, stable temporal behavior is as important as per-frame alignment quality.

Fourth, I include low-rank behavior by reporting rank of the low-rank component, or effective rank, to verify whether RPCA-related objectives are actually achieved.

Finally, I include runtime as an efficiency measure for practical relevance.

For statistical validation, I adopt paired comparisons at the patient level. The primary test is Wilcoxon signed-rank, and paired t-test may be used when normality assumptions are satisfied. I will report p-values, effect sizes, and 95 percent confidence intervals. Because multiple metrics are tested, I will apply Holm-Bonferroni correction to control false positives.

In short, this design allows me to answer not only whether one method is better, but also where it is better and how confident we can be in that conclusion.

Slide 5: Challenges, Timeline, Expected Outcome (~1:20)

There are three main challenges. First, groupwise registration can be computationally expensive. Second, some parameters can be sensitive and may affect stability. Third, intermediate-frame label limitations can constrain direct validation in certain settings.

To mitigate these risks, I will use a fixed reproducible protocol, unified preprocessing, consistent stopping criteria, and robust paired statistics. I will also report variance and failure cases instead of only best-case examples.

My timeline has three milestones. M1 is data pipeline and reproduction of both methods. M2 is metric extraction and statistical testing. M3 is result interpretation, writing, and final presentation.

The expected outcome is a reproducible, statistically grounded evaluation framework and a clear conclusion on whether RPCA enhancement provides real benefit for cardiac cine MRI motion tracking.

Thank you for listening. I welcome your questions and suggestions.